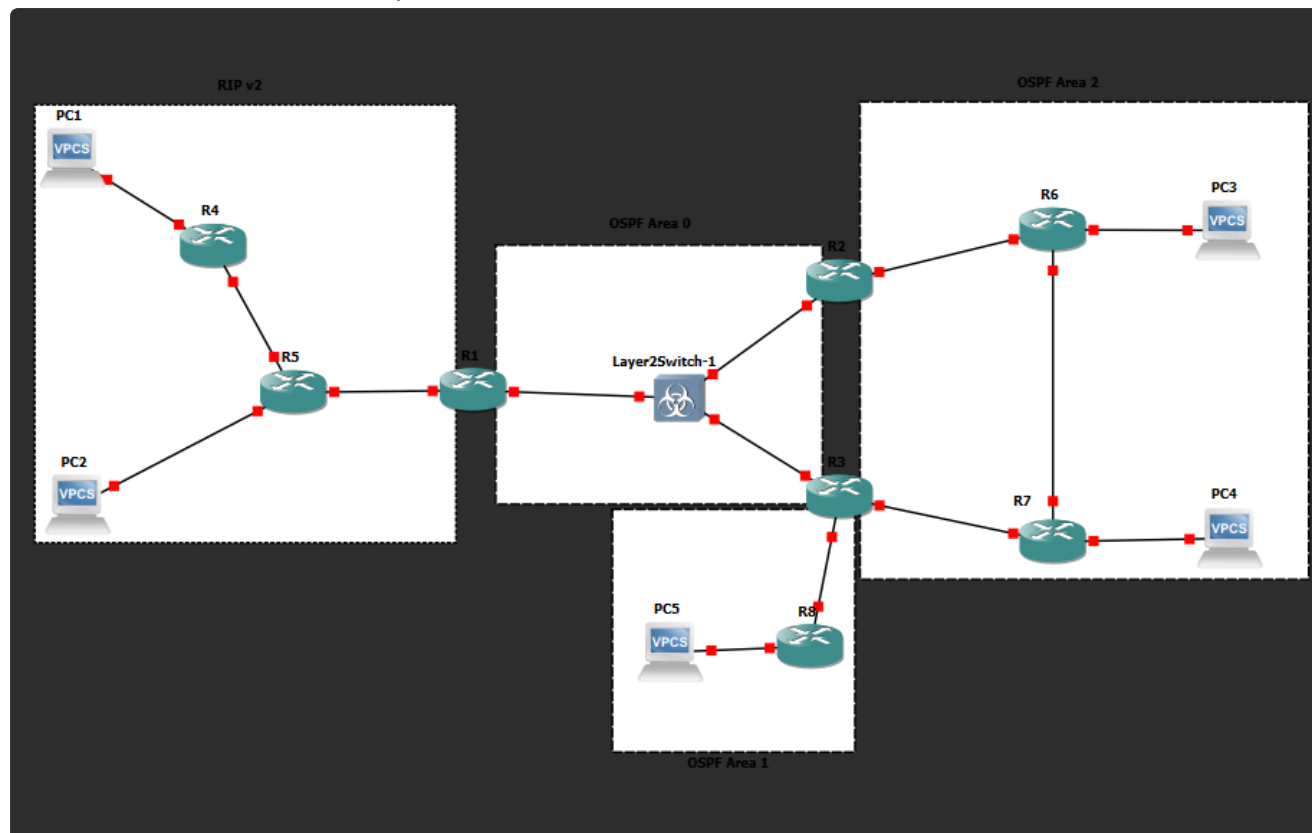


Лабораторная 5

Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"



1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

(nb!) Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.

->

- одна общая сеть R1–R2–R3 через L2-коммутатор;
- шесть соединений маршрутизатор–маршрутизатор;
- пять соединений маршрутизатор–VPC.

Сегмент	Область	Подсеть	Назначение адресов	Broadcast
R1–SW–R2–R3	OSPF Area 0	10.40.5.0/29	R1 .1, R2 .2, R3 .3, SW .4	10.40.5.7
R4–PC1	RIP v2	10.40.5.8/30	R4 .9, PC1 .10	10.40.5.11
R4–R5	RIP v2	10.40.5.12/30	R4 .13, R5 .14	10.40.5.15
R5–PC2	RIP v2	10.40.5.16/30	R5 .17, PC2 .18	10.40.5.19
R5–R1	RIP v2	10.40.5.20/30	R5 .21, R1 .22	10.40.5.23
R2–R6	OSPF Area 2	10.40.5.24/30	R2 .25, R6 .26	10.40.5.27
R6–PC3	OSPF Area 2	10.40.5.28/30	R6 .29, PC3 .30	10.40.5.31
R6–R7	OSPF Area 2	10.40.5.32/30	R6 .33, R7 .34	10.40.5.35
R3–R7	OSPF Area 2	10.40.5.36/30	R3 .37, R7 .38	10.40.5.39
R7–PC4	OSPF Area 2	10.40.5.40/30	R7 .41, PC4 .42	10.40.5.43
R3–R8	OSPF Area 1	10.40.5.44/30	R3 .45, R8 .46	10.40.5.47
R8–PC5	OSPF Area 1	10.40.5.48/30	R8 .49, PC5 .50	10.40.5.51

- Для общей сети R1–R2–R3 достаточно /29, т.к. в ней четыре адресуемых устройства, включая интерфейс управления коммутатором.
- Для остальных соединений достаточно /30: два доступных адреса -> по одному на каждый конец.
- На линиях маршрутизатор-VPC первый адрес получает маршрутизатор, второй - компьютер.

-> Маски

- /29 -> 255.255.255.248 ;
- /30 -> 255.255.255.252 .

Полная таблица адресации:

Устройство	Интерфейс	Сосед	IP-адрес	Маска
R1	Fa0/0	R5 Fa2/0	10.40.5.22/30	255.255.255.252
R1	Fa1/0	SW e0	10.40.5.1/29	255.255.255.248
R2	Fa0/0	SW e1	10.40.5.2/29	255.255.255.248
R2	Fa1/0	R6 Fa0/0	10.40.5.25/30	255.255.255.252
R3	Fa0/0	SW e2	10.40.5.3/29	255.255.255.248
R3	Fa1/0	R7 Fa0/0	10.40.5.37/30	255.255.255.252
R3	Fa2/0	R8 Fa0/0	10.40.5.45/30	255.255.255.252
R4	Fa0/0	PC1	10.40.5.9/30	255.255.255.252

Устройство	Интерфейс	Сосед	IP-адрес	Маска
R4	Fa1/0	R5 Fa1/0	10.40.5.13/30	255.255.255.252
R5	Fa0/0	PC2	10.40.5.17/30	255.255.255.252
R5	Fa1/0	R4 Fa1/0	10.40.5.14/30	255.255.255.252
R5	Fa2/0	R1 Fa0/0	10.40.5.21/30	255.255.255.252
R6	Fa0/0	R2 Fa1/0	10.40.5.26/30	255.255.255.252
R6	Fa1/0	PC3	10.40.5.29/30	255.255.255.252
R6	Fa2/0	R7 Fa2/0	10.40.5.33/30	255.255.255.252
R7	Fa0/0	R3 Fa1/0	10.40.5.38/30	255.255.255.252
R7	Fa1/0	PC4	10.40.5.41/30	255.255.255.252
R7	Fa2/0	R6 Fa2/0	10.40.5.34/30	255.255.255.252
R8	Fa0/0	R3 Fa2/0	10.40.5.46/30	255.255.255.252
R8	Fa1/0	PC5	10.40.5.49/30	255.255.255.252
PC1	e0	R4	10.40.5.10/30	шлюз 10.40.5.9
PC2	e0	R5	10.40.5.18/30	шлюз 10.40.5.17
PC3	e0	R6	10.40.5.30/30	шлюз 10.40.5.29
PC4	e0	R7	10.40.5.42/30	шлюз 10.40.5.41
PC5	e0	R8	10.40.5.50/30	шлюз 10.40.5.49

Настройка R1:

```

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#description TO_R5_F2/0
R1(config-if)#ip address 10.40.5.22 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:02:19.463: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar  1 00:02:20.479: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R1(config)#interface FastEthernet1/0
R1(config-if)#description TO_AREA0_SWITCH
R1(config-if)#ip address 10.40.5.1 255.255.255.248
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar  1 00:03:26.887: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed

```

```
state to up
*Mar  1 00:03:27.887: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R1(config)#end
R1#
*Mar  1 00:03:31.715: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
```

Настройка R2:

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#interface FastEthernet0/0
R2(config-if)#description TO_AREA0_SWITCH
R2(config-if)#ip address 10.40.5.2 255.255.255.248
R2(config-if)#no shutdown
R2(config-if)#e
*Mar  1 00:07:26.671: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar  1 00:07:27.671: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#interface FastEthernet1/0
R2(config-if)#description TO_R6_F0/0
R2(config-if)#ip address 10.40.5.25 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar  1 00:07:49.275: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar  1 00:07:50.275: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R2(config)#end
R2#w
*Mar  1 00:07:53.195: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
```

Настройка R3:

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#hostname R3
R3(config)#interface FastEthernet0/0
```

```
R3(config-if)#description TO_AREA0_SWITCH
R3(config-if)#ip address 10.40.5.3 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:09:26.255: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar 1 00:09:27.255: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R3(config)#interface FastEthernet1/0
R3(config-if)#description TO_R7_F0/0_AREA2
R3(config-if)#ip address 10.40.5.37 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:09:45.915: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar 1 00:09:46.915: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R3(config)#interface FastEthernet2/0
R3(config-if)#description TO_R8_F0/0_AREA1
R3(config-if)#ip address 10.40.5.45 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:10:03.343: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed
state to up
*Mar 1 00:10:04.343: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet2/0, changed state to up
R3(config)#end
R3#w
*Mar 1 00:10:06.119: %SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
[OK]
```

Настройка R4:

```
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#hostname R4
R4(config)#interface FastEthernet0/0
R4(config-if)#description TO_PC1
R4(config-if)#ip address 10.40.5.9 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar 1 00:05:53.967: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
```

```
*Mar 1 00:05:54.967: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R4(config)#interface FastEthernet1/0
R4(config-if)#description TO_R5_F1/0
R4(config-if)#ip address 10.40.5.13 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
*Mar 1 00:06:36.315: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar 1 00:06:37.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R4(config)#end
R4#
*Mar 1 00:06:39.675: %SYS-5-CONFIG_I: Configured from console by console
R4#write memory
Building configuration...
[OK]
```

Настройка R5:

```
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#hostname R5
R5(config)#interface FastEthernet0/0
R5(config-if)#description TO_PC2
R5(config-if)#ip address 10.40.5.17 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
*Mar 1 00:07:54.299: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar 1 00:07:55.299: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R5(config-if)#exit
R5(config)#interface FastEthernet1/0
R5(config-if)#description TO_R4_F1/0
R5(config-if)#ip address 10.40.5.14 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar 1 00:08:19.107: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar 1 00:08:20.107: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R5(config)#interface FastEthernet2/0
R5(config-if)#description TO_R1_F0/0
R5(config-if)#ip address 10.40.5.21 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
```

```
*Mar 1 00:08:44.459: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed
state to up
*Mar 1 00:08:45.459: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet2/0, changed state to up
R5(config)#end
R5#
*Mar 1 00:08:47.535: %SYS-5-CONFIG_I: Configured from console by console
R5#write memory
Building configuration...
[OK]
```

Настройка R6:

```
R6#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#hostname R6
R6(config)#interface FastEthernet0/0
R6(config-if)#description TO_R2_F1/0
R6(config-if)#ip address 10.40.5.26 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
*Mar 1 00:07:34.863: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar 1 00:07:35.863: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R6(config)#interface FastEthernet1/0
R6(config-if)#
R6(config-if)#description TO_PC3
R6(config-if)#ip address 10.40.5.29 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
*Mar 1 00:07:57.407: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar 1 00:07:58.407: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R6(config)#interface FastEthernet2/0
R6(config-if)#description TO_R7_F2/0
R6(config-if)#ip address 10.40.5.33 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
*Mar 1 00:08:14.951: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed
state to up
*Mar 1 00:08:15.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet2/0, changed state to up
R6(config)#end
R6#
```

```
R6#
*Mar  1 00:08:19.263: %SYS-5-CONFIG_I: Configured from console by console
R6#write memory
Building configuration...
[OK]
```

Настройка R7:

```
R7#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#hostname R7
R7(config)#interface FastEthernet0/0
R7(config-if)#description T0_R3_F1/0
R7(config-if)#ip address 10.40.5.38 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Mar  1 00:08:24.203: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar  1 00:08:25.203: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R7(config)#interface FastEthernet1/0
R7(config-if)#description T0_PC4
R7(config-if)#ip address 10.40.5.41 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Mar  1 00:08:48.907: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar  1 00:08:49.907: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R7(config)#interface FastEthernet2/0
R7(config-if)#description T0_R6_F2/0
R7(config-if)#ip address 10.40.5.34 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Mar  1 00:09:30.163: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed
state to up
*Mar  1 00:09:31.163: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet2/0, changed state to up
R7(config)#end
R7#
*Mar  1 00:09:36.207: %SYS-5-CONFIG_I: Configured from console by console
R7#write memory
Building configuration...
[OK]
```


Настройка R8:

```
R8#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#hostname R8
R8(config)#interface FastEthernet0/0
R8(config-if)#description TO_R3_F2/0
R8(config-if)#ip address 10.40.5.46 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
*Mar  1 00:09:31.987: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state to up
*Mar  1 00:09:32.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R8(config)#interface FastEthernet1/0
R8(config-if)#description TO_PC5
R8(config-if)#ip address 10.40.5.49 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
*Mar  1 00:09:49.947: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state to up
*Mar  1 00:09:50.947: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R8(config)#end
R8#
*Mar  1 00:09:54.959: %SYS-5-CONFIG_I: Configured from console by console
R8#write memory
Building configuration...
[OK]
```

Настройка PC1:

```
PC1> ip 10.40.5.10/30 10.40.5.9
Checking for duplicate address...
PC1 : 10.40.5.10 255.255.255.252 gateway 10.40.5.9

PC1> save
Saving startup configuration to startup.vpc
.  done

PC1> show ip

NAME          : PC1[1]
IP/MASK       : 10.40.5.10/30
GATEWAY       : 10.40.5.9
```

```
DNS      :  
MAC      : 00:50:79:66:68:00  
LPORT    : 21240  
RHOST:PORT : 127.0.0.1:21241  
MTU      : 1500
```

```
PC1> ping 10.40.5.9
```

```
84 bytes from 10.40.5.9 icmp_seq=1 ttl=255 time=9.076 ms  
84 bytes from 10.40.5.9 icmp_seq=2 ttl=255 time=5.393 ms  
84 bytes from 10.40.5.9 icmp_seq=3 ttl=255 time=5.224 ms  
84 bytes from 10.40.5.9 icmp_seq=4 ttl=255 time=5.813 ms  
84 bytes from 10.40.5.9 icmp_seq=5 ttl=255 time=5.510 ms
```

Настройка PC2:

```
PC2> ip 10.40.5.18/30 10.40.5.17  
Checking for duplicate address...  
PC2 : 10.40.5.18 255.255.255.252 gateway 10.40.5.17
```

```
PC2> save  
Saving startup configuration to startup.vpc  
. done
```

```
PC2> show ip
```

```
NAME      : PC2[1]  
IP/MASK    : 10.40.5.18/30  
GATEWAY    : 10.40.5.17  
DNS        :  
MAC        : 00:50:79:66:68:01  
LPORT      : 21242  
RHOST:PORT : 127.0.0.1:21243  
MTU        : 1500
```

```
PC2> ping 10.40.5.17
```

```
84 bytes from 10.40.5.17 icmp_seq=1 ttl=255 time=9.457 ms  
84 bytes from 10.40.5.17 icmp_seq=2 ttl=255 time=5.597 ms  
84 bytes from 10.40.5.17 icmp_seq=3 ttl=255 time=5.164 ms  
84 bytes from 10.40.5.17 icmp_seq=4 ttl=255 time=5.961 ms  
84 bytes from 10.40.5.17 icmp_seq=5 ttl=255 time=5.554 ms
```

Настройка P3:

```
PC3> ip 10.40.5.30/30 10.40.5.29  
Checking for duplicate address...  
PC3 : 10.40.5.30 255.255.255.252 gateway 10.40.5.29
```

```
PC3> save
Saving startup configuration to startup.vpc
. done
```

```
PC3> show ip
```

```
NAME       : PC3[1]
IP/MASK    : 10.40.5.30/30
GATEWAY    : 10.40.5.29
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 21244
RHOST:PORT : 127.0.0.1:21245
MTU        : 1500
```

```
PC3> ping 10.40.5.29
```

```
84 bytes from 10.40.5.29 icmp_seq=1 ttl=255 time=9.818 ms
84 bytes from 10.40.5.29 icmp_seq=2 ttl=255 time=5.135 ms
84 bytes from 10.40.5.29 icmp_seq=3 ttl=255 time=5.816 ms
84 bytes from 10.40.5.29 icmp_seq=4 ttl=255 time=5.374 ms
84 bytes from 10.40.5.29 icmp_seq=5 ttl=255 time=5.405 ms
```

Настройка PC4:

```
PC4> ip 10.40.5.42/30 10.40.5.41
Checking for duplicate address...
PC4 : 10.40.5.42 255.255.255.252 gateway 10.40.5.41
```

```
PC4> save
Saving startup configuration to startup.vpc
. done
```

```
PC4> show ip
```

```
NAME       : PC4[1]
IP/MASK    : 10.40.5.42/30
GATEWAY    : 10.40.5.41
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 21246
RHOST:PORT : 127.0.0.1:21247
MTU        : 1500
```

```
PC4> ping 10.40.5.41
```

```
84 bytes from 10.40.5.41 icmp_seq=1 ttl=255 time=9.235 ms
84 bytes from 10.40.5.41 icmp_seq=2 ttl=255 time=5.939 ms
```

```
84 bytes from 10.40.5.41 icmp_seq=3 ttl=255 time=5.694 ms
84 bytes from 10.40.5.41 icmp_seq=4 ttl=255 time=5.975 ms
84 bytes from 10.40.5.41 icmp_seq=5 ttl=255 time=5.934 ms
```

Настройка PC5:

```
PC5> ip 10.40.5.50/30 10.40.5.49
Checking for duplicate address...
PC5 : 10.40.5.50 255.255.255.252 gateway 10.40.5.49
```

```
PC5> save
Saving startup configuration to startup.vpc
. done
```

```
PC5> show ip
```

```
NAME       : PC5[1]
IP/MASK     : 10.40.5.50/30
GATEWAY     : 10.40.5.49
DNS         :
MAC         : 00:50:79:66:68:04
LPORT      : 21248
RHOST:PORT  : 127.0.0.1:21249
MTU         : 1500
```

```
PC5> ping 10.40.5.49
```

```
84 bytes from 10.40.5.49 icmp_seq=1 ttl=255 time=9.438 ms
84 bytes from 10.40.5.49 icmp_seq=2 ttl=255 time=5.877 ms
84 bytes from 10.40.5.49 icmp_seq=3 ttl=255 time=5.325 ms
84 bytes from 10.40.5.49 icmp_seq=4 ttl=255 time=5.666 ms
84 bytes from 10.40.5.49 icmp_seq=5 ttl=255 time=5.624 ms
```

Настройка коммутатора Layer2Switch-1

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#hostname Layer2Switch-1
Layer2Switch-1(config)#interface range GigabitEthernet0/0 - 2
Layer2Switch-1(config-if-range)#switchport mode access
Layer2Switch-1(config-if-range)#switchport access vlan 1
Layer2Switch-1(config-if-range)#no shutdown
Layer2Switch-1(config-if-range)#exit
Layer2Switch-1(config)#interface Vlan1
Layer2Switch-1(config-if)#
*Aug 13 10:01:11.005: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1,
```

```

changed state to down
Layer2Switch-1(config-if)#ip address 10.40.5.4 255.255.255.248
Layer2Switch-1(config-if)#no shutdown
*Aug 13 10:01:22.467: %LINK-3-UPDOWN: Interface Vlan1, changed state to up
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#
*Aug 13 10:01:23.467: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1,
changed state to up
Layer2Switch-1(config)#ip default-gateway 10.40.5.1
Layer2Switch-1(config)#end
Layer2Switch-1#
*Aug 13 10:01:42.086: %SYS-5-CONFIG_I: Configured from console by console
Layer2Switch-1#write memory
Building configuration...
Compressed configuration from 4933 bytes to 1888 bytes[OK]
*Aug 13 10:01:51.148: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updated on disk. Please wait...
*Aug 13 10:01:51.893: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfully.
Layer2Switch-1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	unassigned	YES	unset	up
GigabitEthernet0/1	unassigned	YES	unset	up
GigabitEthernet0/2	unassigned	YES	unset	up
Vlan1	10.40.5.4	YES	manual	up

```

Layer2Switch-1#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

```

Layer2Switch-1#

```

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

Пусть R4 и R5 обмениваются маршрутами, затем R5 передаёт их R1. Интерфейсы к PC и OSPF-части сделал пассивными чтобы их сети объявлялись, но RIP-пакеты туда не шли.

Настройка RIP для R4:

```
R4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#passive-interface default
R4(config-router)#no passive-interface FastEthernet1/0
R4(config-router)#network 10.0.0.0
R4(config-router)#end
R4#
*Mar  1 00:42:53.115: %SYS-5-CONFIG_I: Configured from console by console
R4#write memory
Building configuration...
[OK]
```

Настройка RIP для R5:

```
R5#confi
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#no auto-summary
R5(config-router)#passive-interface default
R5(config-router)#no passive-interface FastEthernet1/0
R5(config-router)#no passive-interface FastEthernet2/0
R5(config-router)#network 10.0.0.0
R5(config-router)#end
R5#
*Mar  1 01:31:57.747: %SYS-5-CONFIG_I: Configured from console by console
R5#write memory
Building configuration...
[OK]
```

Настройка RIP для R1:

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-
R1(config-router)#no auto-summary
R1(config-router)#passive-interface default
R1(config-router)#no passive-
R1(config-router)#no passive-interface FastEthernet0/0
```

```
R1(config-router)#network 10.0.0.0
R1(config-router)#end
R1#
*Mar  1 01:45:15.195: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
```

Проверка процесса RIP:

```
R4#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 15 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface          Send  Recv  Triggered RIP  Key-chain
  FastEthernet1/0      2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Passive Interface(s):
    FastEthernet0/0
    FastEthernet2/0
    VoIP-Null0
  Routing Information Sources:
    Gateway          Distance    Last Update
    10.40.5.14        120         00:00:21
  Distance: (default is 120)

R4#show ip route rip
    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
R       10.40.5.0/29 [120/2] via 10.40.5.14, 00:00:11, FastEthernet1/0
R       10.40.5.20/30 [120/1] via 10.40.5.14, 00:00:11, FastEthernet1/0
R       10.40.5.16/30 [120/1] via 10.40.5.14, 00:00:11, FastEthernet1/0
R4#show ip rip database
10.0.0.0/8    auto-summary
10.40.5.0/29
    [2] via 10.40.5.14, 00:00:15, FastEthernet1/0
10.40.5.8/30   directly connected, FastEthernet0/0
10.40.5.12/30  directly connected, FastEthernet1/0
10.40.5.16/30
    [1] via 10.40.5.14, 00:00:15, FastEthernet1/0
10.40.5.20/30
    [1] via 10.40.5.14, 00:00:15, FastEthernet1/0
```

->

```
R5#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 19 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface          Send  Recv  Triggered RIP  Key-chain
  FastEthernet1/0      2     2
  FastEthernet2/0      2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Passive Interface(s):
    FastEthernet0/0
    VoIP-Null0
  Routing Information Sources:
    Gateway          Distance    Last Update
    10.40.5.13        120         00:00:11
    10.40.5.22        120         00:00:18
  Distance: (default is 120)
```

```
R5#show ip route rip
  10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
R       10.40.5.0/29 [120/1] via 10.40.5.22, 00:00:22, FastEthernet2/0
R       10.40.5.8/30 [120/1] via 10.40.5.13, 00:00:15, FastEthernet1/0
R5#show ip rip database
10.0.0.0/8    auto-summary
10.40.5.0/29
  [1] via 10.40.5.22, 00:00:27, FastEthernet2/0
10.40.5.8/30
  [1] via 10.40.5.13, 00:00:20, FastEthernet1/0
10.40.5.12/30  directly connected, FastEthernet1/0
10.40.5.16/30  directly connected, FastEthernet0/0
10.40.5.20/30  directly connected, FastEthernet2/0
```

->

```
R1#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 22 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
```



```

Redistributing: rip
Default version control: send version 2, receive version 2
  Interface          Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0      2    2
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
Passive Interface(s):
  FastEthernet1/0
  VoIP-Null0
Routing Information Sources:
  Gateway          Distance      Last Update
  10.40.5.21        120           00:00:24
Distance: (default is 120)

```

```

R1#show ip route rip
      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
R      10.40.5.12/30 [120/1] via 10.40.5.21, 00:00:02, FastEthernet0/0
R      10.40.5.8/30 [120/2] via 10.40.5.21, 00:00:02, FastEthernet0/0
R      10.40.5.16/30 [120/1] via 10.40.5.21, 00:00:02, FastEthernet0/0
R1#show ip rip database
10.0.0.0/8      auto-summary
10.40.5.0/29    directly connected, FastEthernet1/0
10.40.5.8/30
      [2] via 10.40.5.21, 00:00:05, FastEthernet0/0
10.40.5.12/30
      [1] via 10.40.5.21, 00:00:05, FastEthernet0/0
10.40.5.16/30
      [1] via 10.40.5.21, 00:00:05, FastEthernet0/0
10.40.5.20/30    directly connected, FastEthernet0/0

```

Проверка PC1 <-> PC2

```
PC1> ping 10.40.5.18
```

```

84 bytes from 10.40.5.18 icmp_seq=1 ttl=62 time=39.689 ms
84 bytes from 10.40.5.18 icmp_seq=2 ttl=62 time=26.087 ms
84 bytes from 10.40.5.18 icmp_seq=3 ttl=62 time=25.542 ms
84 bytes from 10.40.5.18 icmp_seq=4 ttl=62 time=25.459 ms
84 bytes from 10.40.5.18 icmp_seq=5 ttl=62 time=24.998 ms

```

```
PC1>
```

```
PC2> ping 10.40.5.10
```

```

84 bytes from 10.40.5.10 icmp_seq=1 ttl=62 time=22.390 ms
84 bytes from 10.40.5.10 icmp_seq=2 ttl=62 time=25.434 ms
84 bytes from 10.40.5.10 icmp_seq=3 ttl=62 time=25.943 ms

```

```
84 bytes from 10.40.5.10 icmp_seq=4 ttl=62 time=25.614 ms
84 bytes from 10.40.5.10 icmp_seq=5 ttl=62 time=25.782 ms
```

```
PC2>
```

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

Распределение Интерфейсов по областям

Маршрутизатор	Интерфейс	Подсеть	Area
R1	Fa1/0	10.40.5.0/29	0
R2	Fa0/0	10.40.5.0/29	0
R2	Fa1/0	10.40.5.24/30	2
R3	Fa0/0	10.40.5.0/29	0
R3	Fa1/0	10.40.5.36/30	2
R3	Fa2/0	10.40.5.44/30	1
R6	Fa0/0	10.40.5.24/30	2
R6	Fa1/0	10.40.5.28/30	2
R6	Fa2/0	10.40.5.32/30	2
R7	Fa0/0	10.40.5.36/30	2
R7	Fa1/0	10.40.5.40/30	2
R7	Fa2/0	10.40.5.32/30	2
R8	Fa0/0	10.40.5.44/30	1
R8	Fa1/0	10.40.5.48/30	1

Wildcard маски:

- /29 -> 0.0.0.7
- /30 -> 0.0.0.3

Настройка R1 - area 0

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#passive-interface default
R1(config-router)#no passive-interface FastEthernet1/0
```

```
R1(config-router)#network 10.40.5.0 0.0.0.7 area 0
R1(config-router)#end
R1#
*Mar  1 02:03:50.971: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
```

Настройка R2 - area 0 and area 2

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#passive-interface default
R2(config-router)#no passive-interface FastEthernet0/0
R2(config-router)#no passive-interface FastEthernet1/0
R2(config-router)#network 10.40.5.0 0.0.0.7 area 0
R2(config-router)#network 10.40.5.24 0.0.0.3 area 2
R2(config-router)#end
R2#
*Mar  1 02:05:06.703: %SYS-5-CONFIG_I: Configured from console by console
*Mar  1 02:05:07.899: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R2#write memory
Building configuration...
[OK]
```

Настройка R3 - area 0, 1, 2

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#passive-interface default
R3(config-router)#no passive-interface FastEthernet0/0
R3(config-router)#no passive-interface FastEthernet1/0
R3(config-router)#no passive-interface FastEthernet2/0
R3(config-router)#network 10.40.5.0 0.0.0.7 area 0
R3(config-router)#
R3(config-router)#
R3(config-router)#network 10.40.5.36 0.0.0.3 area 2
R3(config-router)#
*Mar  1 02:06:48.963: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done
*Mar  1 02:06:48.983: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R3(config-router)#network 10.40.5.44 0.0.0.3 area 1
```

```
R3(config-router)#area 1 stub no-summary
R3(config-router)#end
R3#
*Mar  1 02:07:12.107: %SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
```

Настройка R6 - area 2

```
R6#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#router ospf 1
R6(config-router)#router-id 6.6.6.6
R6(config-router)#passive-interface default
R6(config-router)#no passive-interface FastEthernet0/0
R6(config-router)#no passive-interface FastEthernet2/0
R6(config-router)#network 10.40.5.24 0.0.0.3 area 2
R6(config-router)#network 10.40.5.28 0.0.0.3 area 2
R6(config-router)#network 10.40.5.32 0.0.0.3 area 2
R6(config-router)#end
*Mar  1 01:58:22.591: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R6(config-router)#end
R6#
*Mar  1 01:58:24.543: %SYS-5-CONFIG_I: Configured from console by console
R6#write memory
Building configuration...
[OK]
```

Настройка R7 - area 2

```
R7#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#passive-interface default
R7(config-router)#no passive-interface FastEthernet0/0
R7(config-router)#no passive-interface FastEthernet2/0
R7(config-router)#network 10.40.5.32 0.0.0.3 area 2
R7(config-router)#network 10.40.5.36 0.0.0.3 area 2
R7(config-router)#
*Mar  1 01:59:11.039: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on
FastEthernet2/0 from LOADING to FULL, Loading Done
R7(config-router)#network 10.40.5.40 0.0.0.3 area 2
R7(config-router)#
*Mar  1 01:59:18.079: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on
FastEthernet0/0 from LOADING to FULL, Loading Done
```

```
R7(config-router)#end
R7#
*Mar  1 01:59:21.087: %SYS-5-CONFIG_I: Configured from console by console
R7#write memory
Building configuration...
[OK]
```

Настройка R8 - area 1 (тупиковая)

```
R8#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#router ospf 1
R8(config-router)#router-id 8.8.8.8
R8(config-router)#passive-interface default
R8(config-router)#no passive-
R8(config-router)#no passive-interface Fasr
R8(config-router)#no passive-interface Fas
R8(config-router)#no passive-interface FastEthernet0/0
R8(config-router)#network 10.40.5.44 0.0.0.3 area 1
R8(config-router)#network 10.40.5.48 0.0.0.3 area 1
R8(config-router)#area 1 stub
R8(config-router)#end
R8#
*Mar  1 01:59:09.971: %SYS-5-CONFIG_I: Configured from console by console
R8#write memory
Building configuration...
[OK]
R8#
*Mar  1 01:59:16.035: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on
FastEthernet0/0 from LOADING to FULL, Loading Done
```

Проверка соседства:

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/BDR	00:00:33	10.40.5.2	FastEthernet1/0
3.3.3.3	1	FULL/DROTHER	00:00:34	10.40.5.3	FastEthernet1/0

```
R1#
```

```
R2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:34	10.40.5.1	FastEthernet0/0

```

3.3.3.3      1    FULL/DROTHER  00:00:36  10.40.5.3
FastEthernet0/0
6.6.6.6      1    FULL/BDR    00:00:31  10.40.5.26
FastEthernet1/0
R2#

```

```
R3#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:35	10.40.5.1	
FastEthernet0/0					
2.2.2.2	1	FULL/BDR	00:00:36	10.40.5.2	
FastEthernet0/0					
8.8.8.8	1	FULL/BDR	00:00:37	10.40.5.46	
FastEthernet2/0					
7.7.7.7	1	FULL/BDR	00:00:30	10.40.5.38	
FastEthernet1/0					

```
R3#
```

```
R6#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
7.7.7.7	1	FULL/BDR	00:00:38	10.40.5.34	
FastEthernet2/0					
2.2.2.2	1	FULL/DR	00:00:32	10.40.5.25	
FastEthernet0/0					

```
R6#
```

```
R7#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/DR	00:00:30	10.40.5.37	
FastEthernet0/0					
6.6.6.6	1	FULL/DR	00:00:33	10.40.5.33	
FastEthernet2/0					

```
R7#
```

```
R8#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/DR	00:00:36	10.40.5.45	
FastEthernet0/0					

Проверка областей:

```
R1#show ip ospf
Routing Process "ospf 1" with ID 1.1.1.1
Start time: 02:02:34.076, Time elapsed: 00:16:27.152
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPF's 10000 msec
Maximum wait time between two consecutive SPF's 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
```

```
R1#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 23 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface          Send  Recv  Triggered RIP  Key-chain
    FastEthernet0/0      2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Passive Interface(s):
    FastEthernet1/0
    VoIP-Null0
  Routing Information Sources:
    Gateway          Distance      Last Update
    10.40.5.21        120          00:00:17
  Distance: (default is 120)
```

```
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
```

```
Incoming update filter list for all interfaces is not set
Router ID 1.1.1.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
  10.40.5.0 0.0.0.7 area 0
Reference bandwidth unit is 100 mbps
Passive Interface(s):
  FastEthernet0/0
  VoIP-Null0
Routing Information Sources:
  Gateway          Distance      Last Update
  3.3.3.3           110           00:06:28
  2.2.2.2           110           00:08:47
Distance: (default is 110)
```

R1#

R1#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Fa1/0	1	0	10.40.5.1/29	1	DR	2/2	

R1#

R2#show ip ospf

```
Routing Process "ospf 1" with ID 2.2.2.2
Start time: 02:04:35.872, Time elapsed: 00:16:08.204
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
It is an area border router
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPF's 10000 msec
Maximum wait time between two consecutive SPF's 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 2. 2 normal 0 stub 0 nssa
Number of areas transit capable is 0
```



```
External flood list length 0
Area BACKBONE(0)
  Number of interfaces in this area is 1
  Area has no authentication
  SPF algorithm last executed 00:14:03.112 ago
  SPF algorithm executed 5 times
  Area ranges are
  Number of LSA 16. Checksum Sum 0x07C069
  Number of opaque link LSA 0. Checksum Sum 0x000000
  Number of DCbitless LSA 0
  Number of indication LSA 0
  Number of DoNotAge LSA 0
  Flood list length 0
```

```
Area 2
  Number of interfaces in this area is 1
  Area has no authentication
  SPF algorithm last executed 00:10:12.592 ago
  SPF algorithm executed 5 times
  Area ranges are
  Number of LSA 13. Checksum Sum 0x08D39A
  Number of opaque link LSA 0. Checksum Sum 0x000000
  Number of DCbitless LSA 0
  Number of indication LSA 0
  Number of DoNotAge LSA 0
  Flood list length 0
```

```
R2#show ip protocols
```

```
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  It is an area border router
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.40.5.0 0.0.0.7 area 0
    10.40.5.24 0.0.0.3 area 2
```

```
Reference bandwidth unit is 100 mbps
```

```
Passive Interface(s):
```

```
VoIP-Null0
```

```
Routing Information Sources:
```

Gateway	Distance	Last Update
7.7.7.7	110	00:10:18
6.6.6.6	110	00:10:18
3.3.3.3	110	00:08:03

```
Distance: (default is 110)
```

```
R2#show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Fa0/0	1	0	10.40.5.2/29	1	BDR	2/2	

Fa1/0	1	2	10.40.5.25/30	1	DR	1/1
R2#						

R3#show ip ospf

```
Routing Process "ospf 1" with ID 3.3.3.3
Start time: 02:06:16.340, Time elapsed: 00:15:44.580
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
It is an area border router
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPF's 10000 msec
Maximum wait time between two consecutive SPF's 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 3. 2 normal 1 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 1
    Area has no authentication
    SPF algorithm last executed 00:14:59.092 ago
    SPF algorithm executed 5 times
    Area ranges are
    Number of LSA 16. Checksum Sum 0x07C069
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
  Area 1
    Number of interfaces in this area is 1
    It is a stub area, no summary LSA in this area
      generates stub default route with cost 1
    Area has no authentication
    SPF algorithm last executed 00:08:55.944 ago
```

SPF algorithm executed 3 times
Area ranges are
Number of LSA 4. Checksum Sum 0x028157
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Area 2

Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 00:11:16.748 ago
SPF algorithm executed 5 times
Area ranges are
Number of LSA 13. Checksum Sum 0x08D39A
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

R3#

R3#show ip protocols

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 3.3.3.3

It is an area border router

Number of areas in this router is 3. 2 normal 1 stub 0 nssa

Maximum path: 4

Routing for Networks:

10.40.5.0 0.0.0.7 area 0

10.40.5.36 0.0.0.3 area 2

10.40.5.44 0.0.0.3 area 1

Reference bandwidth unit is 100 mbps

Passive Interface(s):

VoIP-Null0

Routing Information Sources:

Gateway	Distance	Last Update
8.8.8.8	110	00:09:12
7.7.7.7	110	00:11:30
6.6.6.6	110	00:11:30
2.2.2.2	110	00:11:30

Distance: (default is 110)

R3#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Fa0/0	1	0	10.40.5.3/29	1	DROTH	2/2	
Fa2/0	1	1	10.40.5.45/30	1	DR	1/1	

Fa1/0	1	2	10.40.5.37/30	1	DR	1/1
R3#						

R6#show ip ospf

```
Routing Process "ospf 1" with ID 6.6.6.6
Start time: 01:57:51.244, Time elapsed: 00:14:39.636
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPF's 10000 msec
Maximum wait time between two consecutive SPF's 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
```

Area 2

```
Number of interfaces in this area is 3
Area has no authentication
SPF algorithm last executed 00:12:08.816 ago
SPF algorithm executed 4 times
Area ranges are
Number of LSA 13. Checksum Sum 0x08D39A
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
```

R6#

R6#show ip protocols

Routing Protocol is "ospf 1"

```
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
```

```
Router ID 6.6.6.6
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
  10.40.5.24 0.0.0.3 area 2
  10.40.5.28 0.0.0.3 area 2
  10.40.5.32 0.0.0.3 area 2
Reference bandwidth unit is 100 mbps
Passive Interface(s):
  FastEthernet1/0
  VoIP-Null0
Routing Information Sources:
  Gateway          Distance      Last Update
  7.7.7.7           110           00:12:11
  3.3.3.3           110           00:09:56
  2.2.2.2           110           00:09:56
Distance: (default is 110)
```

```
R6#show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Fa2/0	1	2	10.40.5.33/30	1	DR	1/1	
Fa1/0	1	2	10.40.5.29/30	1	DR	0/0	
Fa0/0	1	2	10.40.5.26/30	1	BDR	1/1	

```
R6#
```

```
R7#show ip ospf
```

```
Routing Process "ospf 1" with ID 7.7.7.7
Start time: 01:58:21.272, Time elapsed: 00:13:43.208
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPF's 10000 msec
Maximum wait time between two consecutive SPF's 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
```

Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0

Area 2

Number of interfaces in this area is 3
Area has no authentication
SPF algorithm last executed 00:12:45.252 ago
SPF algorithm executed 2 times
Area ranges are
Number of LSA 13. Checksum Sum 0x08D39A
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

R7#show ip protocols

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 7.7.7.7

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

10.40.5.32 0.0.0.3 area 2

10.40.5.36 0.0.0.3 area 2

10.40.5.40 0.0.0.3 area 2

Reference bandwidth unit is 100 mbps

Passive Interface(s):

FastEthernet1/0

VoIP-Null0

Routing Information Sources:

Gateway	Distance	Last Update
3.3.3.3	110	00:10:30
6.6.6.6	110	00:12:48
2.2.2.2	110	00:12:48

Distance: (default is 110)

R7#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Fa1/0	1	2	10.40.5.41/30	1	DR	0/0	
Fa0/0	1	2	10.40.5.38/30	1	BDR	1/1	
Fa2/0	1	2	10.40.5.34/30	1	BDR	1/1	

R7#

R8#show ip ospf

Routing Process "ospf 1" with ID 8.8.8.8
Start time: 01:58:06.052, Time elapsed: 00:12:10.772
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPFs 10000 msec
Maximum wait time between two consecutive SPFs 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 0 normal 1 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0

Area 1

Number of interfaces in this area is 2
It is a stub area
Area has no authentication
SPF algorithm last executed 00:10:58.664 ago
SPF algorithm executed 4 times
Area ranges are
Number of LSA 4. Checksum Sum 0x028157
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

R8#show ip protocols

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 8.8.8.8

Number of areas in this router is 1. 0 normal 1 stub 0 nssa

Maximum path: 4

Routing for Networks:

10.40.5.44 0.0.0.3 area 1

10.40.5.48 0.0.0.3 area 1

Reference bandwidth unit is 100 mbps

Passive Interface(s):

FastEthernet1/0

VoIP-Null0

Routing Information Sources:

Gateway	Distance	Last Update
3.3.3.3	110	00:11:01

Distance: (default is 110)

R8#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Fa1/0	1	1	10.40.5.49/30	1	DR	0/0	
Fa0/0	1	1	10.40.5.46/30	1	BDR	1/1	

R8#

Проверка маршрутов

R1#show ip route ospf

10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks

```
O IA    10.40.5.36/30 [110/2] via 10.40.5.3, 00:14:09, FastEthernet1/0
O IA    10.40.5.32/30 [110/3] via 10.40.5.3, 00:14:12, FastEthernet1/0
          [110/3] via 10.40.5.2, 00:14:12, FastEthernet1/0
O IA    10.40.5.44/30 [110/2] via 10.40.5.3, 00:18:02, FastEthernet1/0
O IA    10.40.5.40/30 [110/3] via 10.40.5.3, 00:14:09, FastEthernet1/0
O IA    10.40.5.48/30 [110/3] via 10.40.5.3, 00:11:54, FastEthernet1/0
O IA    10.40.5.28/30 [110/3] via 10.40.5.2, 00:14:12, FastEthernet1/0
O IA    10.40.5.24/30 [110/2] via 10.40.5.2, 00:14:12, FastEthernet1/0
R1#
```

R2#show ip route ospf

10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks

```
O      10.40.5.36/30 [110/3] via 10.40.5.26, 00:14:36, FastEthernet1/0
O      10.40.5.32/30 [110/2] via 10.40.5.26, 00:14:36, FastEthernet1/0
O IA    10.40.5.44/30 [110/2] via 10.40.5.3, 00:14:36, FastEthernet0/0
O      10.40.5.40/30 [110/3] via 10.40.5.26, 00:14:36, FastEthernet1/0
O IA    10.40.5.48/30 [110/3] via 10.40.5.3, 00:12:21, FastEthernet0/0
O      10.40.5.28/30 [110/2] via 10.40.5.26, 00:14:36, FastEthernet1/0
R2#
```

R3#show ip route ospf

10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks

```
O      10.40.5.32/30 [110/2] via 10.40.5.38, 00:14:56, FastEthernet1/0
```



```
0      10.40.5.40/30 [110/2] via 10.40.5.38, 00:14:56, FastEthernet1/0
0      10.40.5.48/30 [110/2] via 10.40.5.46, 00:12:38, FastEthernet2/0
0      10.40.5.28/30 [110/3] via 10.40.5.38, 00:14:56, FastEthernet1/0
0      10.40.5.24/30 [110/3] via 10.40.5.38, 00:14:56, FastEthernet1/0
R3#
```

```
R6#show ip route ospf
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
0      10.40.5.36/30 [110/2] via 10.40.5.34, 00:15:12, FastEthernet2/0
0 IA   10.40.5.44/30 [110/3] via 10.40.5.34, 00:15:12, FastEthernet2/0
      [110/3] via 10.40.5.25, 00:15:12, FastEthernet0/0
0      10.40.5.40/30 [110/2] via 10.40.5.34, 00:15:12, FastEthernet2/0
0 IA   10.40.5.48/30 [110/4] via 10.40.5.34, 00:12:57, FastEthernet2/0
      [110/4] via 10.40.5.25, 00:12:57, FastEthernet0/0
0 IA   10.40.5.0/29 [110/2] via 10.40.5.25, 00:15:12, FastEthernet0/0
R6#
```

```
R7#show ip route ospf
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
0 IA   10.40.5.44/30 [110/2] via 10.40.5.37, 00:15:28, FastEthernet0/0
0 IA   10.40.5.48/30 [110/3] via 10.40.5.37, 00:13:10, FastEthernet0/0
0 IA   10.40.5.0/29 [110/2] via 10.40.5.37, 00:15:28, FastEthernet0/0
0      10.40.5.28/30 [110/2] via 10.40.5.33, 00:15:28, FastEthernet2/0
0      10.40.5.24/30 [110/2] via 10.40.5.33, 00:15:28, FastEthernet2/0
R7#
```

```
R8#show ip route ospf
0*IA 0.0.0.0/0 [110/2] via 10.40.5.45, 00:13:24, FastEthernet0/0
R8#
```

-> R8 не получает отдельный перечень маршрутов Area 0 и Area 2. Для всех внешних направлений он использует маршрут по умолчанию через R3

Проверка связи внутри OSPF:

```
PC3> ping 10.40.5.42
```

```
84 bytes from 10.40.5.42 icmp_seq=1 ttl=62 time=40.143 ms
84 bytes from 10.40.5.42 icmp_seq=2 ttl=62 time=25.696 ms
```

```
84 bytes from 10.40.5.42 icmp_seq=3 ttl=62 time=25.643 ms
84 bytes from 10.40.5.42 icmp_seq=4 ttl=62 time=25.983 ms
84 bytes from 10.40.5.42 icmp_seq=5 ttl=62 time=25.216 ms
```

```
PC3> ping 10.40.5.50
```

```
84 bytes from 10.40.5.50 icmp_seq=1 ttl=60 time=50.827 ms
84 bytes from 10.40.5.50 icmp_seq=2 ttl=60 time=56.785 ms
84 bytes from 10.40.5.50 icmp_seq=3 ttl=60 time=48.129 ms
84 bytes from 10.40.5.50 icmp_seq=4 ttl=60 time=45.606 ms
84 bytes from 10.40.5.50 icmp_seq=5 ttl=60 time=45.906 ms
```

```
PC3>
```

```
PC4> ping 10.40.5.30
```

```
84 bytes from 10.40.5.30 icmp_seq=1 ttl=62 time=24.185 ms
84 bytes from 10.40.5.30 icmp_seq=2 ttl=62 time=25.531 ms
84 bytes from 10.40.5.30 icmp_seq=3 ttl=62 time=26.367 ms
84 bytes from 10.40.5.30 icmp_seq=4 ttl=62 time=25.832 ms
84 bytes from 10.40.5.30 icmp_seq=5 ttl=62 time=25.717 ms
```

```
PC4> ping 10.40.5.50
```

```
84 bytes from 10.40.5.50 icmp_seq=1 ttl=61 time=37.462 ms
84 bytes from 10.40.5.50 icmp_seq=2 ttl=61 time=35.310 ms
84 bytes from 10.40.5.50 icmp_seq=3 ttl=61 time=35.640 ms
84 bytes from 10.40.5.50 icmp_seq=4 ttl=61 time=35.776 ms
84 bytes from 10.40.5.50 icmp_seq=5 ttl=61 time=35.868 ms
```

```
PC4>
```

```
PC5> ping 10.40.5.30
```

```
84 bytes from 10.40.5.30 icmp_seq=1 ttl=60 time=55.361 ms
84 bytes from 10.40.5.30 icmp_seq=2 ttl=60 time=55.588 ms
84 bytes from 10.40.5.30 icmp_seq=3 ttl=60 time=45.866 ms
84 bytes from 10.40.5.30 icmp_seq=4 ttl=60 time=45.338 ms
84 bytes from 10.40.5.30 icmp_seq=5 ttl=60 time=46.022 ms
```

```
PC5> ping 10.40.5.42
```

```
84 bytes from 10.40.5.42 icmp_seq=1 ttl=61 time=39.600 ms
```

```
84 bytes from 10.40.5.42 icmp_seq=2 ttl=61 time=35.847 ms
84 bytes from 10.40.5.42 icmp_seq=3 ttl=61 time=35.910 ms
84 bytes from 10.40.5.42 icmp_seq=4 ttl=61 time=36.521 ms
84 bytes from 10.40.5.42 icmp_seq=5 ttl=61 time=35.942 ms
```

PC5>

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

```
redistribute rip metric 20 metric-type 1 subnets
```

-> импортируем маршруты изученные R1 по RIP в OSPF. Назначаем начальную внешнюю метрику 20, распространяем как O E1(т.к. тут учитывается внешняя метрика + стоимость пути до R1). Сохраняем маски подсетей с помощью subnet.

```
redistribute ospf 1 metric 1
```

-> Берём маршруты процесса OSPF 1 и импортируем их в RIP. Назначаем им начальную RIP-метрику 1.

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#redistribute rip metric 20 metric-type 1 subnets
R1(config-router)#exit
R1(config)#router rip
R1(config-router)#redistribute ospf 1 metric 1
R1(config-router)#end
R1#
*Mar  1 03:59:37.595: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 7 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip, ospf 1
  Default version control: send version 2, receive version 2
    Interface          Send  Recv  Triggered RIP  Key-chain
    FastEthernet0/0      2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Passive Interface(s):
    FastEthernet1/0
    VoIP-Null0
```

Routing Information Sources:

Gateway	Distance	Last Update
10.40.5.21	120	00:00:20

Distance: (default is 120)

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 1.1.1.1

It is an autonomous system boundary router

Redistributing External Routes from,

rip with metric mapped to 20, includes subnets in redistribution

Number of areas in this router is 1. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

10.40.5.0 0.0.0.7 area 0

Reference bandwidth unit is 100 mbps

Passive Interface(s):

FastEthernet0/0

VoIP-Null0

Routing Information Sources:

Gateway	Distance	Last Update
3.3.3.3	110	00:05:28
2.2.2.2	110	00:05:28

Distance: (default is 110)

R1#

R1#show running-config | section router

router ospf 1

router-id 1.1.1.1

log-adjacency-changes

redistribute rip metric 20 metric-type 1 subnets

passive-interface default

no passive-interface FastEthernet1/0

network 10.40.5.0 0.0.0.7 area 0

router rip

version 2

redistribute ospf 1 metric 1

passive-interface default

no passive-interface FastEthernet0/0

network 10.0.0.0

no auto-summary

R1#show ip ospf

Routing Process "ospf 1" with ID 1.1.1.1

Start time: 02:02:34.076, Time elapsed: 02:02:40.412

Supports only single TOS(TOS0) routes

Supports opaque LSA

Supports Link-local Signaling (LLS)

Supports area transit capability

```
It is an autonomous system boundary router
Redistributing External Routes from,
    rip with metric mapped to 20, includes subnets in redistribution
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPF's 10000 msec
Maximum wait time between two consecutive SPF's 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 4. Checksum Sum 0x023039
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
```

```
Area BACKBONE(0)
```

```
    Number of interfaces in this area is 1
    Area has no authentication
    SPF algorithm last executed 00:05:57.500 ago
    SPF algorithm executed 4 times
    Area ranges are
    Number of LSA 16. Checksum Sum 0x096193
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

```
R1#
```

```
R1#
```

```
R1#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static
route
        o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
```

```
0 IA    10.40.5.36/30 [110/2] via 10.40.5.3, 00:06:08, FastEthernet1/0
0 IA    10.40.5.32/30 [110/3] via 10.40.5.3, 00:06:09, FastEthernet1/0
```

```

[110/3] via 10.40.5.2, 00:06:09, FastEthernet1/0
O IA    10.40.5.44/30 [110/2] via 10.40.5.3, 00:06:09, FastEthernet1/0
O IA    10.40.5.40/30 [110/3] via 10.40.5.3, 00:06:09, FastEthernet1/0
O IA    10.40.5.48/30 [110/3] via 10.40.5.3, 00:06:09, FastEthernet1/0
C       10.40.5.0/29 is directly connected, FastEthernet1/0
R       10.40.5.12/30 [120/1] via 10.40.5.21, 00:00:08, FastEthernet0/0
R       10.40.5.8/30 [120/2] via 10.40.5.21, 00:00:08, FastEthernet0/0
C       10.40.5.20/30 is directly connected, FastEthernet0/0
R       10.40.5.16/30 [120/1] via 10.40.5.21, 00:00:08, FastEthernet0/0
O IA    10.40.5.28/30 [110/3] via 10.40.5.2, 00:06:10, FastEthernet1/0
O IA    10.40.5.24/30 [110/2] via 10.40.5.2, 00:06:22, FastEthernet1/0
R1#
R1#show ip ospf database external

```

OSPF Router with ID (1.1.1.1) (Process ID 1)

Type-5 AS External Link States

```

LS age: 388
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.8 (External Network Number )
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0x4898
Length: 36
Network Mask: /30
    Metric Type: 1 (Comparable directly to link state metric)
    TOS: 0
    Metric: 20
    Forward Address: 0.0.0.0
    External Route Tag: 0

```

```

LS age: 388
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.12 (External Network Number )
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0x20BC
Length: 36
Network Mask: /30
    Metric Type: 1 (Comparable directly to link state metric)
    TOS: 0
    Metric: 20
    Forward Address: 0.0.0.0
    External Route Tag: 0

```

```

LS age: 392
Options: (No TOS-capability, DC)

```

LS Type: AS External Link
Link State ID: 10.40.5.16 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0xF7E0
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

LS age: 395
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.20 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0xCF05
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

```
R1#show ip route ospf
    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
O IA    10.40.5.36/30 [110/2] via 10.40.5.3, 00:07:52, FastEthernet1/0
O IA    10.40.5.32/30 [110/3] via 10.40.5.3, 00:07:52, FastEthernet1/0
        [110/3] via 10.40.5.2, 00:07:52, FastEthernet1/0
O IA    10.40.5.44/30 [110/2] via 10.40.5.3, 00:07:52, FastEthernet1/0
O IA    10.40.5.40/30 [110/3] via 10.40.5.3, 00:07:52, FastEthernet1/0
O IA    10.40.5.48/30 [110/3] via 10.40.5.3, 00:07:52, FastEthernet1/0
O IA    10.40.5.28/30 [110/3] via 10.40.5.2, 00:07:52, FastEthernet1/0
O IA    10.40.5.24/30 [110/2] via 10.40.5.2, 00:07:52, FastEthernet1/0
R1#show ip ospf database external
```

OSPF Router with ID (1.1.1.1) (Process ID 1)

Type-5 AS External Link States

LS age: 482
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.8 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001

Checksum: 0x4898
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

LS age: 482
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.12 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0x20BC
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

LS age: 486
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.16 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0xF7E0
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

LS age: 489
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.20 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0xCF05
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0


```
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0
```

```
R1#show ip route rip
    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
R       10.40.5.12/30 [120/1] via 10.40.5.21, 00:00:06, FastEthernet0/0
R       10.40.5.8/30 [120/2] via 10.40.5.21, 00:00:06, FastEthernet0/0
R       10.40.5.16/30 [120/1] via 10.40.5.21, 00:00:06, FastEthernet0/0
R1#
R1#show ip rip database
10.0.0.0/8      auto-summary
10.40.5.0/29    directly connected, FastEthernet1/0
10.40.5.8/30
    [2] via 10.40.5.21, 00:00:11, FastEthernet0/0
10.40.5.12/30
    [1] via 10.40.5.21, 00:00:11, FastEthernet0/0
10.40.5.16/30
    [1] via 10.40.5.21, 00:00:11, FastEthernet0/0
10.40.5.20/30   directly connected, FastEthernet0/0
10.40.5.24/30   redistributed
    [1] via 10.40.5.2, from 2.2.2.2,
10.40.5.28/30   redistributed
    [1] via 10.40.5.2, from 2.2.2.2,
10.40.5.32/30   redistributed
    [1] via 10.40.5.2, from 2.2.2.2,
    [1] via 10.40.5.3, from 3.3.3.3,
10.40.5.36/30   redistributed
    [1] via 10.40.5.3, from 3.3.3.3,
10.40.5.40/30   redistributed
    [1] via 10.40.5.3, from 3.3.3.3,
10.40.5.44/30   redistributed
    [1] via 10.40.5.3, from 3.3.3.3,
10.40.5.48/30   redistributed
    [1] via 10.40.5.3, from 3.3.3.3,
```

->

Редистрибуция на R1 настроилась правильно

- RIP импортирует OSPF - Redistributing: rip, ospf 1
- OSPF импортирует RIP с метрикой 20 и подсетями - Redistributing External Routes from, rip with metric mapped to 20, includes subnets in redistribution
- R1 стал ASBR - It is an autonomous system boundary router
- Созданы четыре внешние LSA типа 5 с Metric Type 1:
10.40.5.8/30
10.40.5.12/30
10.40.5.16/30

10.40.5.20/30

-> Проверка на R2:

```
R2#show ip route ospf
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.40.5.36/30 [110/3] via 10.40.5.26, 00:14:36, FastEthernet1/0
O       10.40.5.32/30 [110/2] via 10.40.5.26, 00:14:36, FastEthernet1/0
O IA    10.40.5.44/30 [110/2] via 10.40.5.3, 00:14:36, FastEthernet0/0
O       10.40.5.40/30 [110/3] via 10.40.5.26, 00:14:36, FastEthernet1/0
O IA    10.40.5.48/30 [110/3] via 10.40.5.3, 00:12:21, FastEthernet0/0
O       10.40.5.28/30 [110/2] via 10.40.5.26, 00:14:36, FastEthernet1/0
R2#
R2#show ip route ospf
      10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
O       10.40.5.36/30 [110/3] via 10.40.5.26, 02:02:10, FastEthernet1/0
O       10.40.5.32/30 [110/2] via 10.40.5.26, 02:02:10, FastEthernet1/0
O IA    10.40.5.44/30 [110/2] via 10.40.5.3, 00:13:21, FastEthernet0/0
O       10.40.5.40/30 [110/3] via 10.40.5.26, 02:02:10, FastEthernet1/0
O IA    10.40.5.48/30 [110/3] via 10.40.5.3, 00:13:21, FastEthernet0/0
O E1    10.40.5.12/30 [110/21] via 10.40.5.1, 00:13:11, FastEthernet0/0
O E1    10.40.5.8/30 [110/21] via 10.40.5.1, 00:13:11, FastEthernet0/0
O E1    10.40.5.20/30 [110/21] via 10.40.5.1, 00:13:11, FastEthernet0/0
O E1    10.40.5.16/30 [110/21] via 10.40.5.1, 00:13:11, FastEthernet0/0
O       10.40.5.28/30 [110/2] via 10.40.5.26, 02:02:10, FastEthernet1/0
R2#show ip ospf database external
```

OSPF Router with ID (2.2.2.2) (Process ID 1)

Type-5 AS External Link States

Routing Bit Set on this LSA

LS age: 812

Options: (No TOS-capability, DC)

LS Type: AS External Link

Link State ID: 10.40.5.8 (External Network Number)

Advertising Router: 1.1.1.1

LS Seq Number: 80000001

Checksum: 0x4898

Length: 36

Network Mask: /30

Metric Type: 1 (Comparable directly to link state metric)

TOS: 0

Metric: 20

Forward Address: 0.0.0.0

External Route Tag: 0

Routing Bit Set on this LSA

LS age: 812

Options: (No TOS-capability, DC)

LS Type: AS External Link
Link State ID: 10.40.5.12 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0x20BC
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

Routing Bit Set on this LSA

LS age: 817
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.16 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0xF7E0
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

Routing Bit Set on this LSA

LS age: 820
Options: (No TOS-capability, DC)
LS Type: AS External Link
Link State ID: 10.40.5.20 (External Network Number)
Advertising Router: 1.1.1.1
LS Seq Number: 80000001
Checksum: 0xCF05
Length: 36
Network Mask: /30
Metric Type: 1 (Comparable directly to link state metric)
TOS: 0
Metric: 20
Forward Address: 0.0.0.0
External Route Tag: 0

-> на R5:

```

R5#show ip route rip
      10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
R       10.40.5.36/30 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R       10.40.5.32/30 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R       10.40.5.44/30 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R       10.40.5.40/30 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R       10.40.5.48/30 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R       10.40.5.0/29 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R       10.40.5.8/30 [120/1] via 10.40.5.13, 00:00:06, FastEthernet1/0
R       10.40.5.28/30 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R       10.40.5.24/30 [120/1] via 10.40.5.22, 00:00:18, FastEthernet2/0
R5#show ip rip database
10.0.0.0/8      auto-summary
10.40.5.0/29
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0
10.40.5.8/30
      [1] via 10.40.5.13, 00:00:10, FastEthernet1/0
10.40.5.12/30    directly connected, FastEthernet1/0
10.40.5.16/30    directly connected, FastEthernet0/0
10.40.5.20/30    directly connected, FastEthernet2/0
10.40.5.24/30
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0
10.40.5.28/30
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0
10.40.5.32/30
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0
10.40.5.36/30
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0
10.40.5.40/30
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0
10.40.5.44/30
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0
10.40.5.48/30
      [1] via 10.40.5.22, 00:00:22, FastEthernet2/0

```

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

Ниже продублированы успешные проверки связи между VPC, которые уже были выполнены ранее при проверке RIP и OSPF в пунктах 2 и 3.

PC1 -> PC2 (проверено ранее в пункте 2)

```

PC1> ping 10.40.5.18

84 bytes from 10.40.5.18 icmp_seq=1 ttl=62 time=39.689 ms

```

```
84 bytes from 10.40.5.18 icmp_seq=2 ttl=62 time=26.087 ms
84 bytes from 10.40.5.18 icmp_seq=3 ttl=62 time=25.542 ms
84 bytes from 10.40.5.18 icmp_seq=4 ttl=62 time=25.459 ms
84 bytes from 10.40.5.18 icmp_seq=5 ttl=62 time=24.998 ms
```

PC2 -> PC1 (проверено ранее в пункте 2)

```
PC2> ping 10.40.5.10

84 bytes from 10.40.5.10 icmp_seq=1 ttl=62 time=22.390 ms
84 bytes from 10.40.5.10 icmp_seq=2 ttl=62 time=25.434 ms
84 bytes from 10.40.5.10 icmp_seq=3 ttl=62 time=25.943 ms
84 bytes from 10.40.5.10 icmp_seq=4 ttl=62 time=25.614 ms
84 bytes from 10.40.5.10 icmp_seq=5 ttl=62 time=25.782 ms
```

PC3 -> PC4 (проверено ранее в пункте 3)

```
PC3> ping 10.40.5.42

84 bytes from 10.40.5.42 icmp_seq=1 ttl=62 time=40.143 ms
84 bytes from 10.40.5.42 icmp_seq=2 ttl=62 time=25.696 ms
84 bytes from 10.40.5.42 icmp_seq=3 ttl=62 time=25.643 ms
84 bytes from 10.40.5.42 icmp_seq=4 ttl=62 time=25.983 ms
84 bytes from 10.40.5.42 icmp_seq=5 ttl=62 time=25.216 ms
```

PC4 -> PC3 (проверено ранее в пункте 3)

```
PC4> ping 10.40.5.30

84 bytes from 10.40.5.30 icmp_seq=1 ttl=62 time=24.185 ms
84 bytes from 10.40.5.30 icmp_seq=2 ttl=62 time=25.531 ms
84 bytes from 10.40.5.30 icmp_seq=3 ttl=62 time=26.367 ms
84 bytes from 10.40.5.30 icmp_seq=4 ttl=62 time=25.832 ms
84 bytes from 10.40.5.30 icmp_seq=5 ttl=62 time=25.717 ms
```

PC3 -> PC5 (проверено ранее в пункте 3)

```
PC3> ping 10.40.5.50

84 bytes from 10.40.5.50 icmp_seq=1 ttl=60 time=50.827 ms
84 bytes from 10.40.5.50 icmp_seq=2 ttl=60 time=56.785 ms
84 bytes from 10.40.5.50 icmp_seq=3 ttl=60 time=48.129 ms
84 bytes from 10.40.5.50 icmp_seq=4 ttl=60 time=45.606 ms
84 bytes from 10.40.5.50 icmp_seq=5 ttl=60 time=45.906 ms
```

PC5 -> PC3 (проверено ранее в пункте 3)

```
PC5> ping 10.40.5.30
```

```
84 bytes from 10.40.5.30 icmp_seq=1 ttl=60 time=55.361 ms
84 bytes from 10.40.5.30 icmp_seq=2 ttl=60 time=55.588 ms
84 bytes from 10.40.5.30 icmp_seq=3 ttl=60 time=45.866 ms
84 bytes from 10.40.5.30 icmp_seq=4 ttl=60 time=45.338 ms
84 bytes from 10.40.5.30 icmp_seq=5 ttl=60 time=46.022 ms
```

PC4 -> PC5 (проверено ранее в пункте 3)

```
PC4> ping 10.40.5.50
```

```
84 bytes from 10.40.5.50 icmp_seq=1 ttl=61 time=37.462 ms
84 bytes from 10.40.5.50 icmp_seq=2 ttl=61 time=35.310 ms
84 bytes from 10.40.5.50 icmp_seq=3 ttl=61 time=35.640 ms
84 bytes from 10.40.5.50 icmp_seq=4 ttl=61 time=35.776 ms
84 bytes from 10.40.5.50 icmp_seq=5 ttl=61 time=35.868 ms
```

PC5 -> PC4 (проверено ранее в пункте 3)

```
PC5> ping 10.40.5.42
```

```
84 bytes from 10.40.5.42 icmp_seq=1 ttl=61 time=39.600 ms
84 bytes from 10.40.5.42 icmp_seq=2 ttl=61 time=35.847 ms
84 bytes from 10.40.5.42 icmp_seq=3 ttl=61 time=35.910 ms
84 bytes from 10.40.5.42 icmp_seq=4 ttl=61 time=36.521 ms
84 bytes from 10.40.5.42 icmp_seq=5 ttl=61 time=35.942 ms
```

Проверка связи между RIP- и OSPF-частями после редистрибуции

PC1 -> PC3

```
PC1> ping 10.40.5.30
```

```
84 bytes from 10.40.5.30 icmp_seq=1 ttl=59 time=70.209 ms
84 bytes from 10.40.5.30 icmp_seq=2 ttl=59 time=56.458 ms
84 bytes from 10.40.5.30 icmp_seq=3 ttl=59 time=55.974 ms
84 bytes from 10.40.5.30 icmp_seq=4 ttl=59 time=55.457 ms
84 bytes from 10.40.5.30 icmp_seq=5 ttl=59 time=55.765 ms
```

PC1 -> PC4

```
PC1> ping 10.40.5.42
```

```
84 bytes from 10.40.5.42 icmp_seq=1 ttl=59 time=62.747 ms
84 bytes from 10.40.5.42 icmp_seq=2 ttl=59 time=55.696 ms
84 bytes from 10.40.5.42 icmp_seq=3 ttl=59 time=55.940 ms
84 bytes from 10.40.5.42 icmp_seq=4 ttl=59 time=56.124 ms
84 bytes from 10.40.5.42 icmp_seq=5 ttl=59 time=56.298 ms
```

PC1 -> PC5

```
PC1> ping 10.40.5.50
```

```
84 bytes from 10.40.5.50 icmp_seq=1 ttl=59 time=63.608 ms
84 bytes from 10.40.5.50 icmp_seq=2 ttl=59 time=55.379 ms
84 bytes from 10.40.5.50 icmp_seq=3 ttl=59 time=55.135 ms
84 bytes from 10.40.5.50 icmp_seq=4 ttl=59 time=55.684 ms
84 bytes from 10.40.5.50 icmp_seq=5 ttl=59 time=55.800 ms
```

PC2 -> PC3

```
PC2> ping 10.40.5.30
```

```
84 bytes from 10.40.5.30 icmp_seq=1 ttl=60 time=60.023 ms
84 bytes from 10.40.5.30 icmp_seq=2 ttl=60 time=45.716 ms
84 bytes from 10.40.5.30 icmp_seq=3 ttl=60 time=46.855 ms
84 bytes from 10.40.5.30 icmp_seq=4 ttl=60 time=46.068 ms
84 bytes from 10.40.5.30 icmp_seq=5 ttl=60 time=45.469 ms
```

PC2 -> PC4

```
PC2> ping 10.40.5.42
```

```
84 bytes from 10.40.5.42 icmp_seq=1 ttl=60 time=41.352 ms
84 bytes from 10.40.5.42 icmp_seq=2 ttl=60 time=44.946 ms
84 bytes from 10.40.5.42 icmp_seq=3 ttl=60 time=46.053 ms
84 bytes from 10.40.5.42 icmp_seq=4 ttl=60 time=45.773 ms
84 bytes from 10.40.5.42 icmp_seq=5 ttl=60 time=46.559 ms
```

PC2 -> PC5

```
PC2> ping 10.40.5.50
```

```
84 bytes from 10.40.5.50 icmp_seq=1 ttl=60 time=47.985 ms
84 bytes from 10.40.5.50 icmp_seq=2 ttl=60 time=46.536 ms
84 bytes from 10.40.5.50 icmp_seq=3 ttl=60 time=56.309 ms
```

```
84 bytes from 10.40.5.50 icmp_seq=4 ttl=60 time=45.771 ms
84 bytes from 10.40.5.50 icmp_seq=5 ttl=60 time=46.160 ms
```

PC3 -> PC1

```
PC3> ping 10.40.5.10
```

```
84 bytes from 10.40.5.10 icmp_seq=1 ttl=59 time=69.721 ms
84 bytes from 10.40.5.10 icmp_seq=2 ttl=59 time=66.191 ms
84 bytes from 10.40.5.10 icmp_seq=3 ttl=59 time=66.569 ms
84 bytes from 10.40.5.10 icmp_seq=4 ttl=59 time=65.915 ms
84 bytes from 10.40.5.10 icmp_seq=5 ttl=59 time=65.709 ms
```

PC3 -> PC2

```
PC3> ping 10.40.5.18
```

```
84 bytes from 10.40.5.18 icmp_seq=1 ttl=60 time=42.793 ms
84 bytes from 10.40.5.18 icmp_seq=2 ttl=60 time=45.712 ms
84 bytes from 10.40.5.18 icmp_seq=3 ttl=60 time=45.441 ms
84 bytes from 10.40.5.18 icmp_seq=4 ttl=60 time=45.826 ms
84 bytes from 10.40.5.18 icmp_seq=5 ttl=60 time=55.476 ms
```

PC4 -> PC1

```
PC4> ping 10.40.5.10
```

```
84 bytes from 10.40.5.10 icmp_seq=1 ttl=59 time=67.630 ms
84 bytes from 10.40.5.10 icmp_seq=2 ttl=59 time=66.086 ms
84 bytes from 10.40.5.10 icmp_seq=3 ttl=59 time=65.570 ms
84 bytes from 10.40.5.10 icmp_seq=4 ttl=59 time=55.738 ms
84 bytes from 10.40.5.10 icmp_seq=5 ttl=59 time=56.070 ms
```

PC4 -> PC2

```
PC4> ping 10.40.5.18
```

```
84 bytes from 10.40.5.18 icmp_seq=1 ttl=60 time=46.558 ms
84 bytes from 10.40.5.18 icmp_seq=2 ttl=60 time=46.091 ms
84 bytes from 10.40.5.18 icmp_seq=3 ttl=60 time=56.297 ms
84 bytes from 10.40.5.18 icmp_seq=4 ttl=60 time=45.265 ms
84 bytes from 10.40.5.18 icmp_seq=5 ttl=60 time=56.068 ms
```

PC5 -> PC1


```
PC5> ping 10.40.5.10
```

```
84 bytes from 10.40.5.10 icmp_seq=1 ttl=59 time=70.481 ms
84 bytes from 10.40.5.10 icmp_seq=2 ttl=59 time=55.673 ms
84 bytes from 10.40.5.10 icmp_seq=3 ttl=59 time=55.242 ms
84 bytes from 10.40.5.10 icmp_seq=4 ttl=59 time=55.222 ms
84 bytes from 10.40.5.10 icmp_seq=5 ttl=59 time=55.326 ms
```

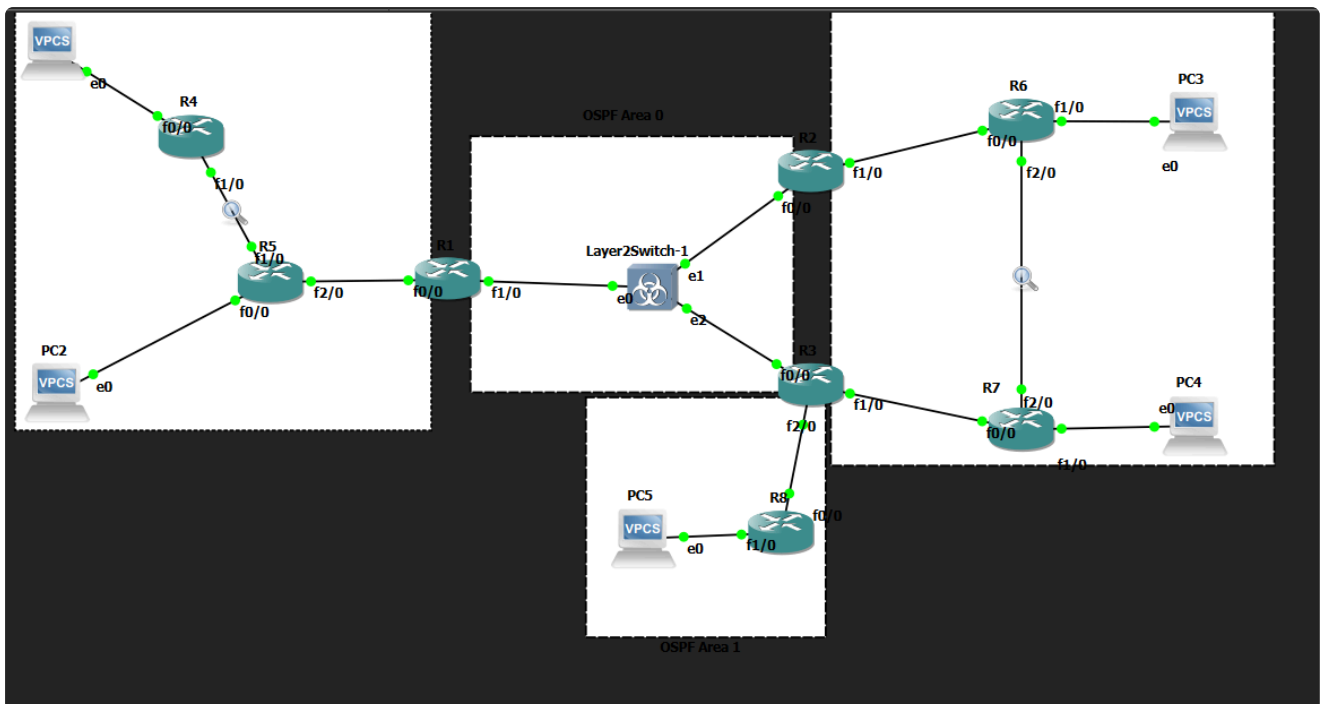
PC5 -> PC2

```
PC5> ping 10.40.5.18
```

```
84 bytes from 10.40.5.18 icmp_seq=1 ttl=60 time=46.493 ms
84 bytes from 10.40.5.18 icmp_seq=2 ttl=60 time=45.524 ms
84 bytes from 10.40.5.18 icmp_seq=3 ttl=60 time=45.089 ms
84 bytes from 10.40.5.18 icmp_seq=4 ttl=60 time=46.352 ms
84 bytes from 10.40.5.18 icmp_seq=5 ttl=60 time=45.966 ms
```

Выполнены все 20 направленных проверок между пятью VPC: каждая из 10 пар проверена в обоих направлениях. Во всех случаях получены ответы на пять ICMP Echo Request без потерь. Следовательно, маршрутизация внутри доменов RIP и OSPF, а также взаимная редистрибуция маршрутов между ними работают корректно.

6) Перехватить в Wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.



*Standard input [R6 FastEthernet2/0 to R7 FastEthernet2/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

ospf

No.	Time	Source	Destination	Protocol	Length	Info
3	0.875664	10.40.5.33	224.0.0.5	OSPF	94	Hello Packet
4	0.937939	10.40.5.34	224.0.0.5	OSPF	94	Hello Packet
7	10.868192	10.40.5.33	224.0.0.5	OSPF	94	Hello Packet
8	10.938502	10.40.5.34	224.0.0.5	OSPF	94	Hello Packet
11	20.868386	10.40.5.33	10.40.5.34	OSPF	78	DB Description
12	20.868416	10.40.5.33	224.0.0.5	OSPF	94	Hello Packet
13	20.930487	10.40.5.34	10.40.5.33	OSPF	78	DB Description
14	20.930511	10.40.5.34	224.0.0.5	OSPF	94	Hello Packet
15	20.938932	10.40.5.33	10.40.5.34	OSPF	138	DB Description
16	20.940520	10.40.5.34	10.40.5.33	OSPF	98	DB Description
17	20.949017	10.40.5.33	10.40.5.34	OSPF	78	DB Description
18	20.949045	10.40.5.33	224.0.0.5	OSPF	98	LS Update
19	20.950575	10.40.5.34	10.40.5.33	OSPF	78	DB Description
20	20.950593	10.40.5.34	10.40.5.33	OSPF	94	LS Request
21	20.950600	10.40.5.34	224.0.0.5	OSPF	98	LS Update
22	20.950668	10.40.5.33	10.40.5.34	OSPF	70	LS Request
23	20.950809	10.40.5.33	10.40.5.34	OSPF	78	DB Description
24	20.950896	10.40.5.33	10.40.5.34	OSPF	186	LS Update
25	20.960675	10.40.5.34	10.40.5.33	OSPF	122	LS Update
26	20.960688	10.40.5.34	10.40.5.33	OSPF	78	LS Acknowledge
27	21.011029	10.40.5.34	224.0.0.5	OSPF	242	LS Update
28	21.019436	10.40.5.33	10.40.5.34	OSPF	98	LS Acknowledge
29	21.212310	10.40.5.34	224.0.0.5	OSPF	98	LS Update
30	21.371930	10.40.5.33	224.0.0.5	OSPF	122	LS Update
31	21.373374	10.40.5.34	224.0.0.5	OSPF	122	LS Update
32	21.412150	10.40.5.33	224.0.0.5	OSPF	94	LS Update
33	21.413629	10.40.5.34	224.0.0.5	OSPF	94	LS Update
34	21.453942	10.40.5.34	224.0.0.5	OSPF	186	LS Update
35	23.466093	10.40.5.34	224.0.0.5	OSPF	158	LS Acknowledge
36	23.475442	10.40.5.33	224.0.0.5	OSPF	258	LS Acknowledge
37	25.071003	10.40.5.34	224.0.0.5	OSPF	98	LS Update

Frame 3: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
 Ethernet II, Src: cc:06:30:97:00:20 (cc:06:30:97:00:20), Dst: IPv4mcast 05 (01:00:5e:00:00:05)
 wireshark_Standard inputONU1T3.pcapng Пакеты: 114 - Отображено: 72 (63.2%) - Потеряно: 0 (0.0%) Профили: Default

*Standard input [R4 FastEthernet1/0 to R5 FastEthernet1/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

rip

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.40.5.14	224.0.0.9	RIPv2	106	Response
3	0.001012	10.40.5.13	224.0.0.9	RIPv2	66	Response
9	59.555820	10.40.5.14	224.0.0.9	RIPv2	106	Response
11	64.334819	10.40.5.13	224.0.0.9	RIPv2	66	Response
13	77.198218	10.40.5.14	224.0.0.9	RIPv2	186	Response
18	126.940867	10.40.5.14	224.0.0.9	RIPv2	246	Response
19	128.279462	10.40.5.13	224.0.0.9	RIPv2	66	Response
28	192.617125	10.40.5.14	224.0.0.9	RIPv2	246	Response
29	197.690983	10.40.5.13	224.0.0.9	RIPv2	66	Response

Frame 1: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface -, id 0
 Ethernet II, Src: cc:05:30:7b:00:10 (cc:05:30:7b:00:10), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
 Internet Protocol Version 4, Src: 10.40.5.14, Dst: 224.0.0.9
 User Datagram Protocol, Src Port: 520, Dst Port: 520
 Routing Information Protocol

RIP v2

*Standard input [R4 FastEthernet1/0 to R5 FastEthernet1/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

rip

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.40.5.14	224.0.0.9	RIPv2	106	Response
3	0.001012	10.40.5.13	224.0.0.9	RIPv2	66	Response
9	59.555820	10.40.5.14	224.0.0.9	RIPv2	106	Response
11	64.334819	10.40.5.13	224.0.0.9	RIPv2	66	Response
13	77.198218	10.40.5.14	224.0.0.9	RIPv2	186	Response
18	126.940867	10.40.5.14	224.0.0.9	RIPv2	246	Response
19	128.279462	10.40.5.13	224.0.0.9	RIPv2	66	Response
28	192.617125	10.40.5.14	224.0.0.9	RIPv2	246	Response
29	197.690983	10.40.5.13	224.0.0.9	RIPv2	66	Response

Frame 1: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface -, id 0

Ethernet II, Src: cc:05:30:7b:00:10 (cc:05:30:7b:00:10), Dst: IPv4mcast_09 (01:00:5e:00:00:09)

Internet Protocol Version 4, Src: 10.40.5.14, Dst: 224.0.0.9

User Datagram Protocol, Src Port: 520, Dst Port: 520

Routing Information Protocol

Command: Response (2)

Version: RIPv2 (2)

- IP Address: 10.40.5.0, Metric: 2
 - Address Family: IP (2)
 - Route Tag: 0
 - IP Address: 10.40.5.0
 - Netmask: 255.255.255.248
 - Next Hop: 0.0.0.0
 - Metric: 2
- IP Address: 10.40.5.16, Metric: 1
 - Address Family: IP (2)
 - Route Tag: 0
 - IP Address: 10.40.5.16
 - Netmask: 255.255.255.252
 - Next Hop: 0.0.0.0
 - Metric: 1
- IP Address: 10.40.5.20, Metric: 1
 - Address Family: IP (2)
 - Route Tag: 0
 - IP Address: 10.40.5.20
 - Netmask: 255.255.255.252
 - Next Hop: 0.0.0.0
 - Metric: 1

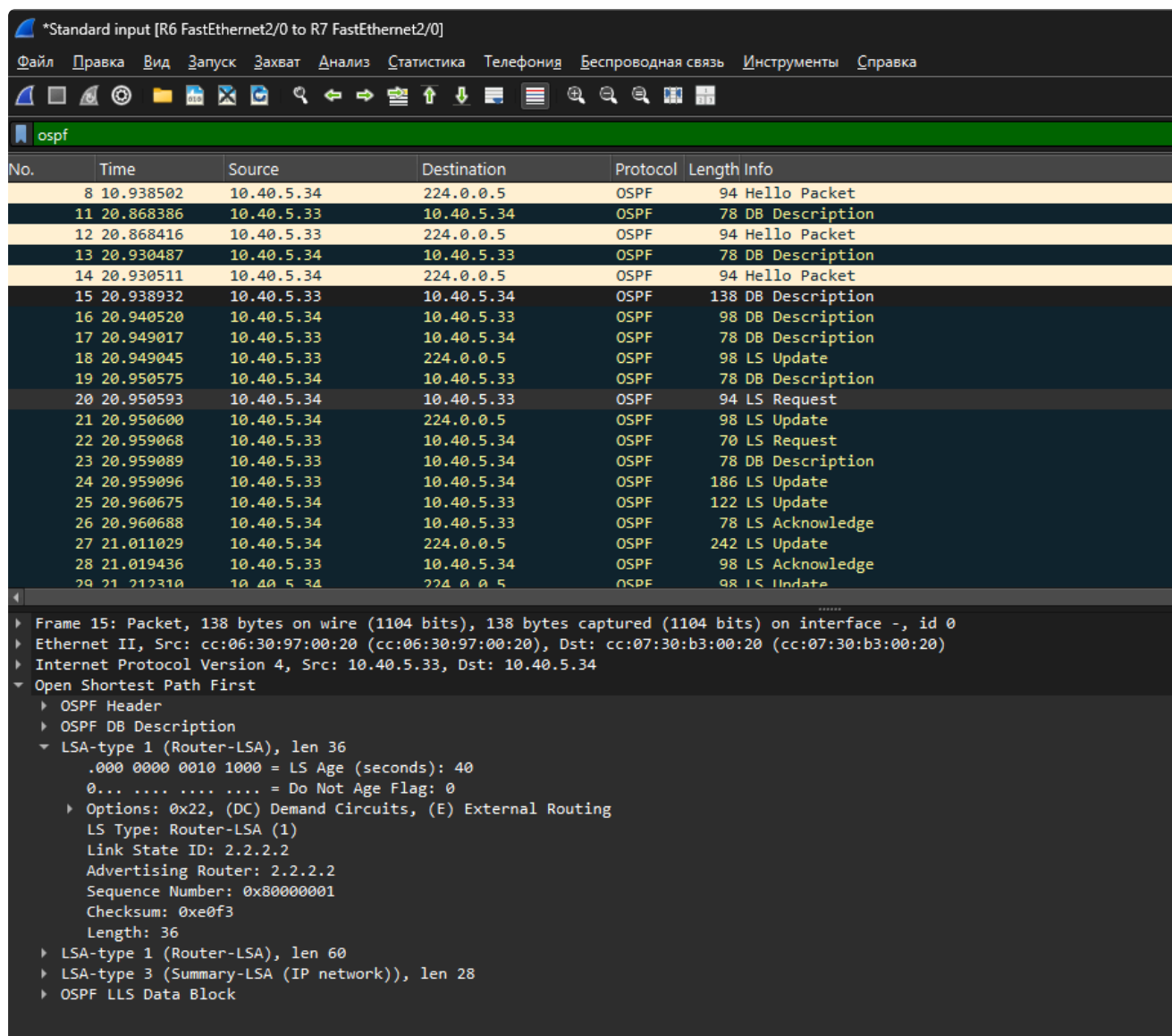
->10.40.5.13 - R4, 10.40.5.14 - R5

получатель 224.0.0.9 — multicast-адрес RIPv2, **тип сообщения Response**, UDP-порт источника и назначения - 520

Внутри Routing Information Protocol можно посмотреть следующую информацию:

- IP Address - адрес сети
 - Netmask - маска сети
 - Next Hop - следующий маршрутизатор
 - Metric - RIP метрика, количество переходов
- Разный размер пакетов (66 , 106 , 186 , 246 байт) означает разное количество маршрутных записей внутри.
- 10.40.5.0, маска 255.255.255.248, метрика 2
 - 10.40.5.16, маска 255.255.255.252, метрика 1
 - 10.40.5.20, маска 255.255.255.252, метрика 1
 - Next Hop = 0.0.0.0 означает, что следующим переходом следует считать отправителя RIP-пакета — R5 (10.40.5.14)

OSPF



Тип	На снимке	Назначение
Hello Packet	№ ..., 8, 12, 14	Обнаружение соседей и поддержание соседства
DB Description	№11, 13, 15–17, ...	Краткое описание содержимого LSDB
LS Request	№20, 22, ...	Запрос недостающих LSA
LS Update	№18, 21, 24–25, ...	Передача самих LSA
LS Acknowledge	№26, 28, ...	Подтверждение получения LSA

Адреса соответствуют каналу R6–R7:

- 10.40.5.33 - R6
- 10.40.5.34 - R7

- 224.0.0.5 - multicast AllSPFRouters

No.	Time	Source	Destination	Protocol	Length	Info
3	0.875664	10.40.5.33	224.0.0.5	OSPF	94	Hello Packet

```

Frame 3: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
Ethernet II, Src: cc:06:30:97:00:20 (cc:06:30:97:00:20), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
Internet Protocol Version 4, Src: 10.40.5.33, Dst: 224.0.0.5
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
  Total Length: 80
  Identification: 0x0007 (7)
  0000 .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 1
  Protocol: OSPF IGP (89)
  Header Checksum: 0xc940 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.40.5.33
  Destination Address: 224.0.0.5
  [Stream index: 0]
Open Shortest Path First
  OSPF Header
    Version: 2
    Message Type: Hello Packet (1)
    Packet Length: 48
    Source OSPF Router: 6.6.6.6
    Area ID: 0.0.0.2
    Checksum: 0xd182 [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  OSPF Hello Packet
  OSPF LLS Data Block
  
```

Видим, что OSPF работает непосредственно поверх IP, используя IP-протокол №89, без TCP и UDP.

У Hello пакета видим корректные Router ID - 6.6.6.6 и Area ID - 0.0.0.2

```

  OSPF Hello Packet
    Network Mask: 255.255.255.252
    Hello Interval [sec]: 10
    Options: 0x12, (L) LLS Data block, (E) External Routing
    Router Priority: 1
    Router Dead Interval [sec]: 40
    Designated Router: 0.0.0.0
    Backup Designated Router: 0.0.0.0
    Active Neighbor: 7.7.7.7
  OSPF LLS Data Block
    Checksum: 0xffff6
    LLS Data Length: 12 bytes
    Extended options TLV
      TLV Type: 1
      TLV Length: 4
      Options: 0x00000001, (LR) LSDB Resynchronization
  
```

Маска, Hello interval - 10 sec, DR - 0.0.0.0 и BDR - 0.0.0.0

сосед - 7.7.7.7

Hello : OSPFv2, Router ID 6.6.6.6 , Area 0.0.0.2 , маска 255.255.255.252 , Hello Interval 10 , Dead Interval 40 , Priority 1 , сосед 7.7.7.7 .

DR = 0.0.0.0 и BDR = 0.0.0.0 : этот Hello был перехвачен в начале установления соседства, до завершения выбора DR/BDR. Это не постоянное состояние сети.

ospf

No.	Time	Source	Destination	Protocol	Length	Info
8	10.938502	10.40.5.34	224.0.0.5	OSPF	94	Hello Packet
11	20.868386	10.40.5.33	10.40.5.34	OSPF	78	DB Description

Frame 11: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface -, id 0
Ethernet II, Src: cc:06:30:97:00:20 (cc:06:30:97:00:20), Dst: cc:07:30:b3:00:20 (cc:07:30:b3:00:20)
Internet Protocol Version 4, Src: 10.40.5.33, Dst: 10.40.5.34
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
Total Length: 64
Identification: 0x000c (12)
000. = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: OSPF IGP (89)
Header Checksum: 0x9a07 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.40.5.33
Destination Address: 10.40.5.34
[Stream index: 2]
Open Shortest Path First
OSPF Header
Version: 2
Message Type: DB Description (2)
Packet Length: 32
Source OSPF Router: 6.6.6.6
Area ID: 0.0.0.2
Checksum: 0x925c [correct]
Instance ID: Base IPv4 Unicast Instance (0)
Auth Type: Null (0)
Auth Data (none): 0000000000000000
OSPF DB Description
Interface MTU: 1500
Options: 0x52, (O) Opaque, (L) LLS Data block, (E) External Routing
DB Description: 0x07, (I) Init, (M) More, (MS) Master
DD Sequence: 1936
OSPF LLS Data Block
Checksum: 0xffff6
LLS Data Length: 12 bytes
Extended options TLV
TLV Type: 1
TLV Length: 4
Options: 0x00000001, (LR) LSDB Resynchronization

-> у DB Description можно посмотреть флаги

DB Description : MTU 1500 , флаги I , M , MS , DD Sequence 1936 ; в другом DBD видны заголовки Router-LSA и Summary-LSA.

```
20 20.950593 10.40.5.34 10.40.5.33 OSPF 94 LS Request

Frame 20: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
Ethernet II, Src: cc:07:30:b3:00:20 (cc:07:30:b3:00:20), Dst: cc:06:30:97:00:20 (cc:06:30:97:00:20)
Internet Protocol Version 4, Src: 10.40.5.34, Dst: 10.40.5.33
Open Shortest Path First
  ▾ OSPF Header
    Version: 2
    Message Type: LS Request (3)
    Packet Length: 60
    Source OSPF Router: 7.7.7.7
    Area ID: 0.0.0.2
    Checksum: 0xbc5f [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  ▾ Link State Request
    LS Type: Router-LSA (1)
    Link State ID: 2.2.2.2
    Advertising Router: 2.2.2.2
  ▾ Link State Request
    LS Type: Router-LSA (1)
    Link State ID: 6.6.6.6
    Advertising Router: 6.6.6.6
  ▾ Link State Request
    LS Type: Summary-LSA (IP network) (3)
    Link State ID: 10.40.5.0
    Advertising Router: 2.2.2.2
```

-> LS Request содержит информацию о том, какие LSA запрашиваются.

LS Request : R7 запрашивает Router-LSA для 2.2.2.2 и 6.6.6.6 , а также Summary-LSA сети 10.40.5.0 , созданную R2.

```
21 20.950600    10.40.5.34      224.0.0.5      OSPF           98 LS Update
<
> Frame 21: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: cc:07:30:b3:00:20 (cc:07:30:b3:00:20), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
> Internet Protocol Version 4, Src: 10.40.5.34, Dst: 224.0.0.5
> Open Shortest Path First
  > OSPF Header
    > Version: 2
    > Message Type: LS Update (4)
    > Packet Length: 64
    > Source OSPF Router: 7.7.7.7
    > Area ID: 0.0.0.2
    > Checksum: 0x2572 [correct]
    > Instance ID: Base IPv4 Unicast Instance (0)
    > Auth Type: Null (0)
    > Auth Data (none): 0000000000000000
  > LS Update Packet
    > Number of LSAs: 1
    > LSA-type 1 (Router-LSA), len 36
      .000 0000 0000 0011 = LS Age (seconds): 3
      0... .... .... = Do Not Age Flag: 0
      > Options: 0x22, (DC) Demand Circuits, (E) External Routing
        0... .... = DN: Not set
        .0... .... = (O) Opaque: Not set
        ..1. .... = (DC) Demand Circuits: Supported
        ...0 .... = (L) LLS Data block: Not Present
        .... 0... = (N) NSSA: Not supported
        .... .0.. = (MC) Multicast: Not capable
        .... .1. = (E) External Routing: Capable
        .... ...0 = (MT) Multi-Topology Routing: No
      LS Type: Router-LSA (1)
      Link State ID: 2.2.2.2
      Advertising Router: 2.2.2.2
      Sequence Number: 0x80000002
      Checksum: 0xfe80
      Length: 36
      > Flags: 0x01, (B) Area border router
        0... .... = (H) Host: No
        ..0. .... = (S) Shortcut-capable ABR: No
        ...0 .... = (N) NSSA translation: No
        .... 0... = (W) Wild-card multicast receiver: No
        .... .0.. = (V) Virtual link endpoint: No
        .... .0. = (E) AS boundary router: No
        .... ...1 = (B) Area border router: Yes
      Number of Links: 1
      > Type: Transit ID: 10.40.5.26      Data: 10.40.5.25      Metric: 1
        Link ID: 10.40.5.26 - IP address of Designated Router
        Link Data: 10.40.5.25
        Link Type: 2 - Connection to a transit network
        Number of Metrics: 0 - TOS
        0 Metric: 1
```

-> LS Update содержит информацию о количестве и содержании передаваемых LSA
LS Update : содержит одну Router-LSA от 2.2.2.2 ; флаг B показывает, что R2 является ABR, стоимость соединения — 1 .


```
26 20.960688      10.40.5.34      10.40.5.33      OSPF      78 LS Acknowledge
<
Frame 26: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface -, id 0
Ethernet II, Src: cc:07:30:b3:00:20 (cc:07:30:b3:00:20), Dst: cc:06:30:97:00:20 (cc:06:30:97:00:20)
Internet Protocol Version 4, Src: 10.40.5.34, Dst: 10.40.5.33
Open Shortest Path First
  OSPF Header
    Version: 2
    Message Type: LS Acknowledge (5)
    Packet Length: 44
    Source OSPF Router: 7.7.7.7
    Area ID: 0.0.0.2
    Checksum: 0x470c [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  LSA-type 1 (Router-LSA), len 36
    .000 0000 0000 0010 = LS Age (seconds): 2
    0... .... .... .... = Do Not Age Flag: 0
  Options: 0x22, (DC) Demand Circuits, (E) External Routing
    0... .... = DN: Not set
    .0.. .... = (O) Opaque: Not set
    ..1. .... = (DC) Demand Circuits: Supported
    ...0 .... = (L) LLS Data block: Not Present
    .... 0... = (N) NSSA: Not supported
    .... .0.. = (MC) Multicast: Not capable
    .... ..1. = (E) External Routing: Capable
    .... ...0 = (MT) Multi-Topology Routing: No
  LS Type: Router-LSA (1)
  Link State ID: 2.2.2.2
  Advertising Router: 2.2.2.2
  Sequence Number: 0x80000002
  Checksum: 0xfe80
  Length: 36
```

-> LS Acknowledge содержит заголовки подтвержденных LSA.

LS Acknowledge : подтверждается Router-LSA от 2.2.2.2 с sequence number 0x80000002 .

7) Сохранить в отдельные файлы с префиксом `rt_` и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

сохранённые файлы в папке `rt`

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.

сохранённые файлы в папке `config`